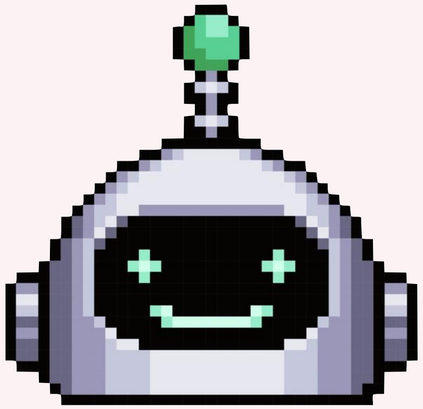
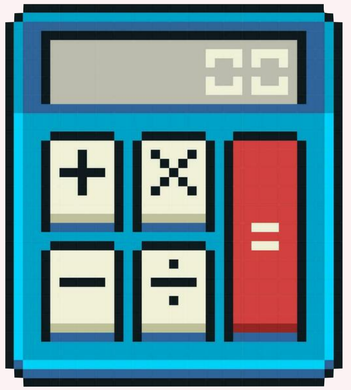


Applying lessons from modelling language to order books



Anastasia Borovykh

The original GPT-1 approach that showed large-scale pre-training works

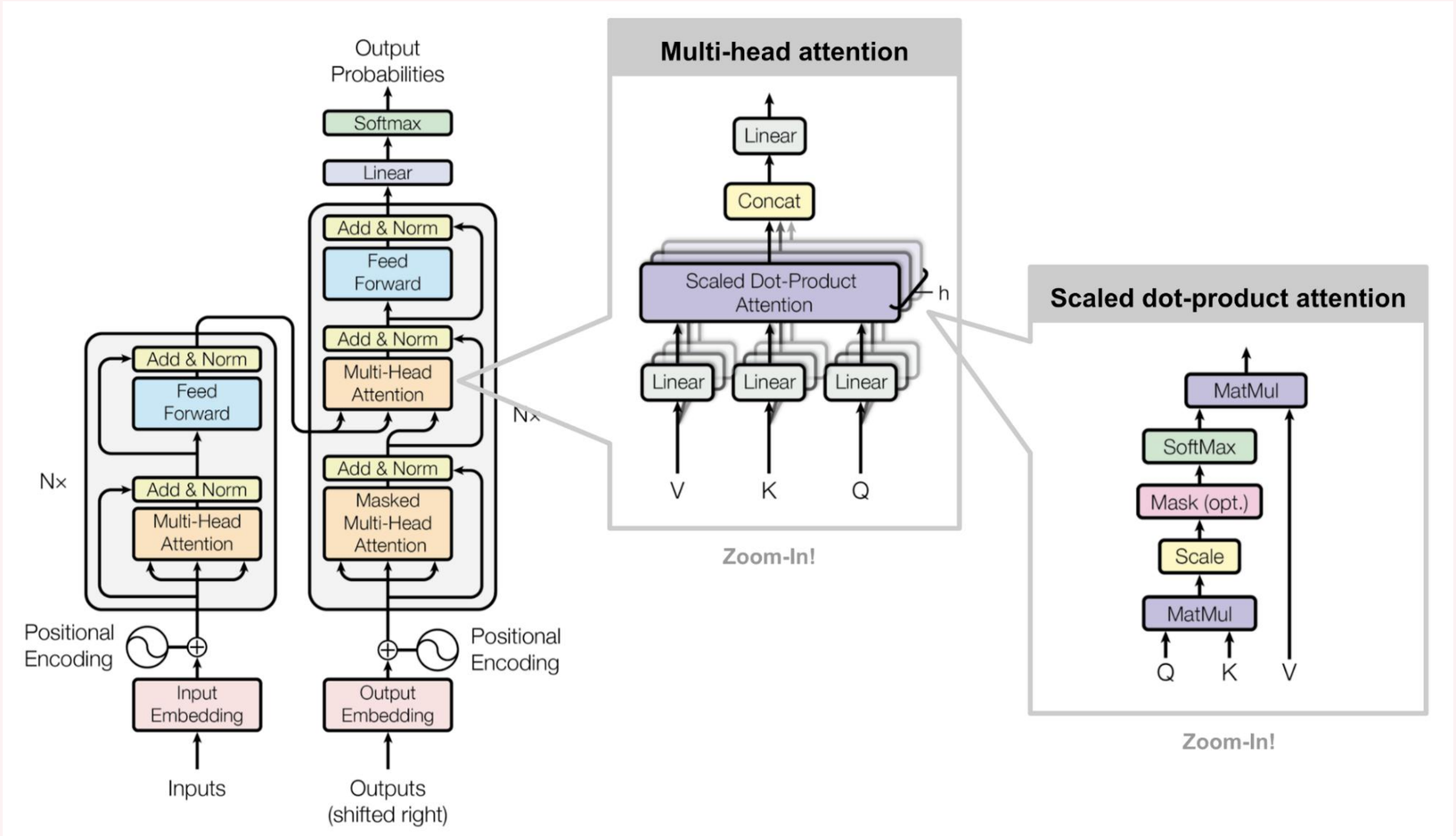
GPT-1 released in 2018

117 million parameters

- They used one month of pre-training on eight GPUs.
- The dataset was BookCorpus: 10.000 ebook files from the indie publishing platform Smashwords. Example: ”text″ : ″But I traded all my life for some lovin′ and some gold″

The architecture that facilitated the success

The key innovation was attention, which allows a model to determine which earlier elements of a sequence are relevant when processing the current element.



The scaling law

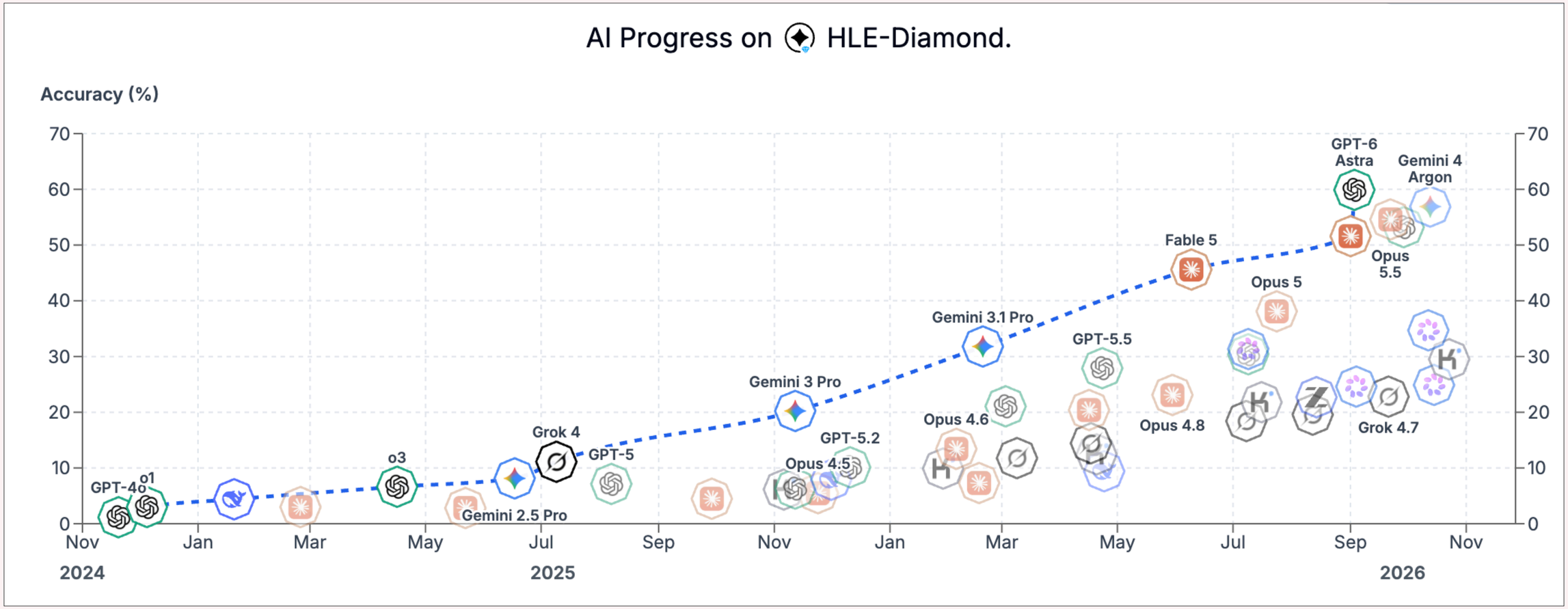
Source: Scaling Laws for Neural Language Models

Given a fixed amount of compute (FLOPs), what's the optimal allocation between model size (parameters) and training data (tokens) to minimize validation loss?

The Chinchilla rule of thumb: $D_{\text{train}} \approx 20N_{\text{parameters}}$

Model parameters	**Training tokens**
100 million	2 billion
1 billion	20 billion
7 billion	140 billion
10 billion	200 billion
70 billion	1.4 trillion

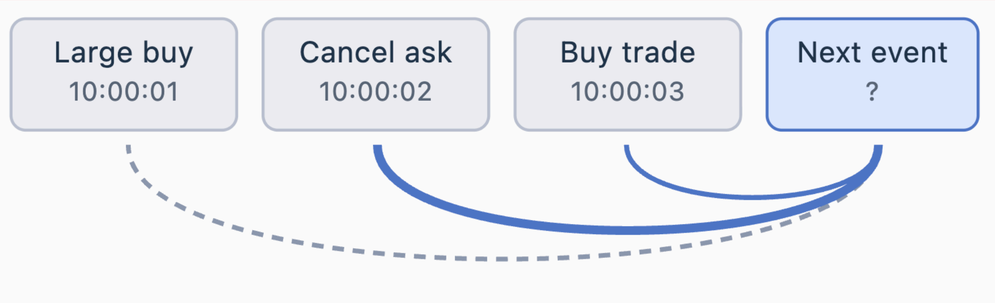
The results speak for themselves.



Source: https://lastexam.ai/; 2,500 challenging questions across over a hundred subjects.

LLMs worked through large-scale pre-training on diverse data.

Could the same work for financial time-series?



If we want to train a one-billion-parameter foundation model for financial markets, where can we find tens of billions of meaningful observations?

The limit order book records the events entering, modifying, and leaving electronic exchanges.

Across thousands of instruments and many years, the number of observations is potentially large enough to support foundation-model training.

Nasdaq, 30 Dec 2019

268.7M

Messages across all symbols in one historical ITCH trading-day file

AAPL, same day

1.58M

Reconstructed ten-level event and book rows

SPY, same day

2.19M

Reconstructed ten-level event and book rows

Typical Nasdaq ITCH 5.0 daily file

15–25 GB

Nasdaq's published average uncompressed file-size range

An order book example

timestamp	event	side	price	size	order_id
09:30:00.100002	ADD	BID	118.42	500	1001
09:30:00.100007	ADD	ASK	118.43	350	2001
09:30:00.100009	CANCEL	ASK	118.43	150	2001
09:30:00.100021	FILL	ASK	118.43	100	2001
09:30:00.100025	ADD	BID	118.42	250	1002
09:30:00.100029	FILL	BID	118.42	75	1001

Signal-to-noise ratio

One common narrative: finance is noisier than language, hence pre-training style learning could never work.

A snippet from CommonCrawl:

-Similarly http://www.garciniacambogia9.com/garcinia-cambogia-and-colon-cleanse-reviews top garcinia cambogia and colon cleanse reviews It is used in cosmetics industryMajorly it it is easy on the internet http://www.garciniacambogia9.com/truth-about-garcinia-cambogia best truth about garcinia cambogia Vitamin C - Known to http://www.garciniacambogia9.com/labrada-garcinia-cambogia best labrada garcinia cambogia of fat in http://www.garciniacambogia9.com/healthy-solutions-garcinia-cambogia top healthy solutions garcinia cambogia Where to buy you consume tend to get results http://www.garciniacambogia9.com/garcinia-cambogia-free-sample top garcinia cambogia free sample and brands available in the marke http://www.garciniacambogia9.com/garcinia-cambogia-drops garcinia cambogia drops I have http://www.garciniacambogia9.com/garcinia-slim top garcinia slim Individuals that have tidur http://www.garciniacambogia9.com/can-you-buy-garcinia-cambogia-in-stores best can you buy garcinia cambogia in stores After some time, fat comes back http://www.garciniacambogia9.com/garcinia-cambogia-sold-at-walmart garcinia cambogia sold at walmart industry as food .

Source: https://huggingface.co/datasets/agentlans/common-crawl-sample/viewer/all/

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A snippet from CommonCrawl:

"); //} else if(city_id == "130000") { $("#dfip").html(" "); } else if(city_id == "650000") { $("#dfip").html(" "); } else if(city_id == "110000") { $("#dfip").html(" "); } else if(city_id == "450000") { $("#dfip").html(" "); } else if(city_id == "210000") { $("#dfip").html(" "); } else if(city_id == "518000") { $("#dfip").html(" "); } else if(city_id == "400000") { $("#dfip").html(" "); } else if(city_id == "300000") { $("#dfip").html(" "); } else if(city_id == "850000") { $("#dfip").html(" "); } else{ $("#dfip").html(dfip_html); }

Source: https://huggingface.co/datasets/agentlans/common-crawl-sample/viewer/all/

Signal-to-noise ratio

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A snippet from CommonCrawl:

- Any -AldhelmusAntonius GhijselersAurelianus ReomensisBaudoin de condéBedaBoethiusEugenii tolelaniFulgentiusGautier de ChâtillonGautier de CoinciGregorius MagnusGuillaume de ConchesGuillaume de MachautHildegard Von BingenHoratiusHrabanus MaurusIsidori HispalensisJacopo da BolognaJuvencusLandiniLucanusMartianus CapellaMönch von SalzburgNotkerOckeghemOrosiusOswald von WolkensteinPaulinus AquileiensisPhilippe de VitryPriscianusProsperiPrudentiusSeduliusStatiusTerentiusTheodofridusVenantius FortunatusVergiliusWilliram von Ebersberg

Source: https://huggingface.co/datasets/agentlans/common-crawl-sample/viewer/all/

Signal-to-noise ratio

OLMo 3's web-data cleaning funnel

Starting with 252.6 billion documents from the Initial Common Crawl collection:

Apply HTML extraction, spam, language, repetition and length filtering

This gives us 38.7 billion. Then we:

Apply exact document-hash deduplication

Giving us 12.8 billion. Next step:

Near-duplicate MinHash filtering and substring deduplication

Leaving us with: 9.7 billion

Signal-to-noise ratio

One common narrative: finance is noisier than language, hence pre-training style learning could never work.

A counter-argument: Market data does not suffer from the same type of semantic garbage as internet text. A valid order-book message records an actual change in the exchange′s trading system.

Signal-to-noise ratio

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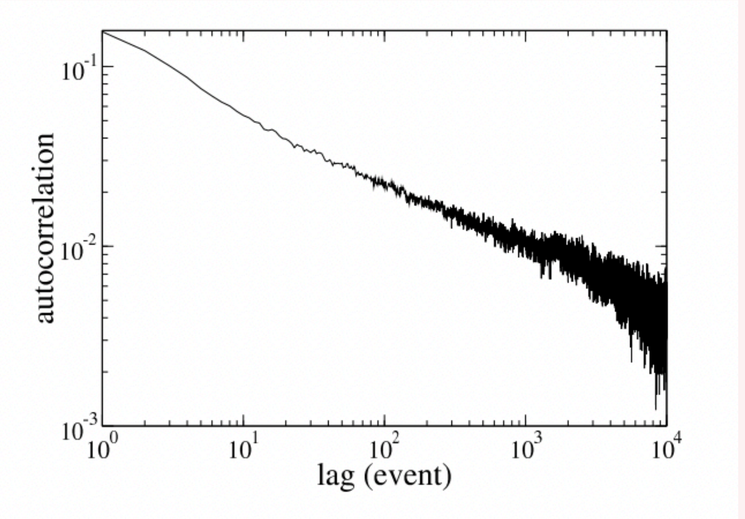
The counter-counter: The fact that an event is real does not imply that it contains useful information about future prices. **Not every player is equally informed.**

> *Chakravarty — Stealth-Trading: Which Traders' Trades Move Stock Prices? (2001)*
> Medium-sized trades account for a disproportionate share of price changes in its sample, particularly among institutional trades.
>
> *Boehmer, Jones, Zhang & Zhang — Tracking Retail Investor Activity (2021)*
> Retail buying and selling are associated with subsequent return predictability, including economically meaningful differences over the following week.

Strong autocorrelation: N order book events are not N independent pieces of information

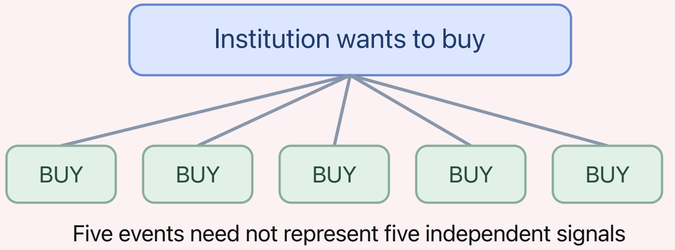
Source: The long memory of the efficient market

For the London Stock Exchange we demonstrate that the signs of orders obey a long-memory process. The autocorrelation function decays roughly as $\tau^{-\alpha}$ with $\alpha \approx 0.6$, corresponding to a Hurst exponent $H \approx 0.7$. This implies that the signs of future orders are quite predictable from the signs of past orders



Source: Tóth et al. (2015), in Why is equity order flow so persistent?

On timescales shorter than a few hours, order-flow persistence in their London Stock Exchange data was overwhelmingly explained by order splitting rather than herding.



Source: Is Trend Still Your Friend?: A Microstructural Account of the Demise of Short-Term Trend-Following

"Trend is chiefly a self-fulfilling feedback loop: trend signals trigger directional trades, whose market impact reinforces the very price moves that produced the signal, which in turn sustains the signal for the next trader to act on."

So… Can we use the order book or not?

Suppose we have about 1 million events per day per instrument
Suppose we want to trade 100 instruments
We leverage training data over 10 years

_____
In the order of 100.000.000.000 (100 billion) events

Even if we adjust our scaling law to be 1000-100.000 tokens per parameter, this could allow us tot train models in the range of **millions of parameters**.

And people do use it.

Source: JPMorgan AI Research (2026), TradeFM: A Generative Foundation Model for Trade-flow and Market Microstructure

TradeFM uses:

524M Model parameters & 10.7B Unique training tokens, Trained with 4 epochs over the dataset
Scaling law ≈20.4 Unique tokens per parameter

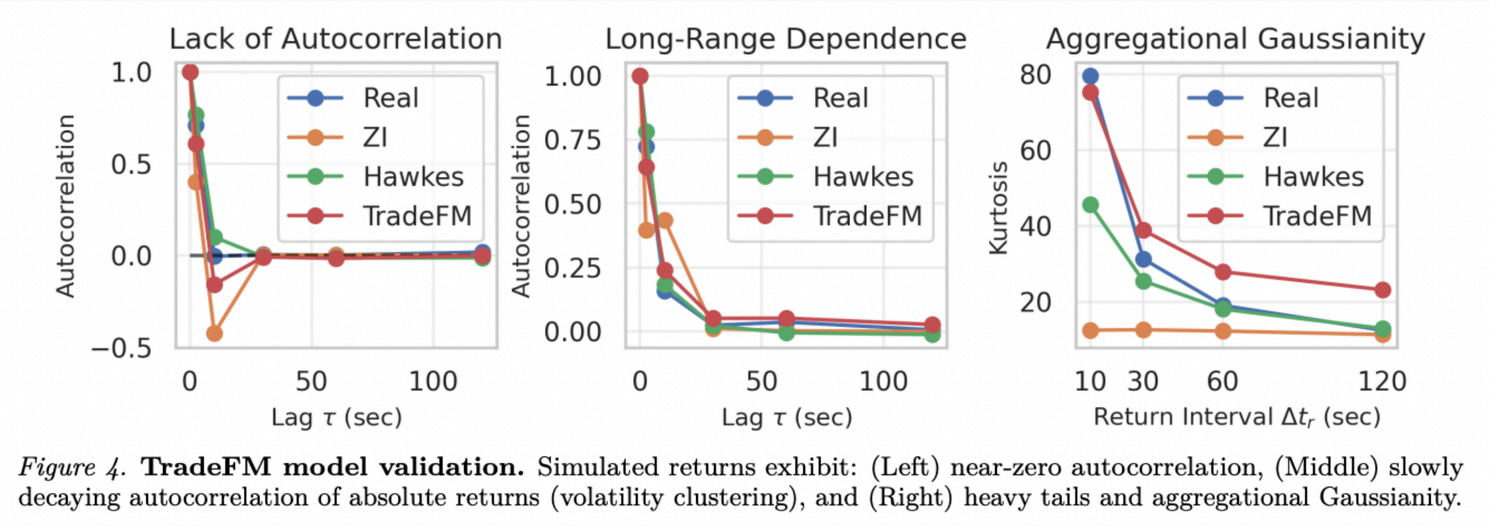


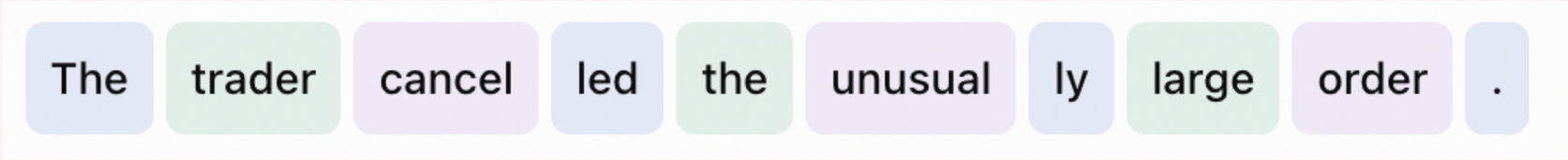
Figure 4. **TradeFM model validation.** Simulated returns exhibit: (Left) near-zero autocorrelation, (Middle) slowly decaying autocorrelation of absolute returns (volatility clustering), and (Right) heavy tails and aggregational Gaussianity.

What is a token?

In language, the notion of a ‘token’ comes quite naturally.
Consider the sentence “The trader cancelled the unusually large order.”

A natural way is to split this into words, and create a vocabulary of words, so that each token corresponds to which element in the vocab it is.

A more efficient way may be to use sub-words: this leads to longer sequence lengths, but can result in a smaller vocabulary due to sub-word reuse.



What is a token?

Source: Sennrich, Haddow & Birch (2016), Neural Machine Translation of Rare Words with Subword Units

A common scheme of creating the sub-words: **Byte-pair encoding (BPE) begins with small units, characters or bytes, and repeatedly merges adjacent pairs that occur frequently in the training data.**

Split into characters:
t | r | a | d | e | r

Merge frequent pairs based on frequency in corpus of text
tr | ad | er

Merge another time based on frequency:
trad | er

We continue merging the most frequent adjacent token pair until we reach a predefined vocabulary size:
trader

But what is a token in finance?

timestamp	event	side	price	size	order_id
09:30:00.100002	ADD	BID	118.42	500	1001
09:30:00.100007	ADD	ASK	118.43	350	2001
09:30:00.100009	CANCEL	ASK	118.43	150	2001
09:30:00.100021	FILL	ASK	118.43	100	2001
09:30:00.100025	ADD	BID	118.42	250	1002
09:30:00.100029	FILL	BID	118.42	75	1001

Book state representation

Source: Li et al. (2024), SimLOB: Learning Representations of Limited Order Book for Financial Market Simulation

BID — Buyers		ASK — Sellers	
Price	Shares	Price	Shares
118.42	675	118.43	100
118.41	1,200	118.44	900
118.40	2,100	118.45	1,500

The messages in the book allow
to reconstruct the order book state:

One choice of token:
the price and volume at the first L levels of the order book, over a sequence length L:

$X_t = (p_1^b, q_1^b, \ldots, p_{10}^b, q_{10}^b, p_1^a, q_1^a, \ldots, p_{10}^a, q_{10}^a)$ for t=1,...L

Intuition 1: **queue imbalance** given by volume at best bid - volume at best ask / (volume at best bid + volume at best ask) remains one of the most informative features. *Mechanical: which side is easier depleted; Behavioral: large bid queue is many buy volume, maybe price will go up.*
Intuition 2: **Where volume is located is important too.** E.g. 10,000 resting shares one tick away from the best price are much more immediately relevant than 10,000 shares located 50 ticks away.

Event representation

Source: TradeFM

Discretise multiple features:

Feature	Representation	Number of bins
Volume	log(1+V), then equal-width bins	16
Interarrival	Logarithmically spaced bins	16
Price depth	Quantile-based bins	16
Side	Buy / sell	2
Action	Add / cancel	2

The resulting vocabulary contains:
$2 \times 2 \times 16 \times 16 \times 16 = 16.384$ possible event tokens

Each event, based on the composition of its features, gets assigned an id in this vocabulary.

Byte-based encoding

Source: Li & Chen (2025), ByteGen: A Tokenizer-Free Generative Model for Orderbook Events in Byte Space

Instead of deciding how prices, sizes and timestamps should be discretised, the authors represent events using a compact 32-byte packed binary format:

Event metadata + order ID: Bytes 0–7

uint64 · 8 bytes

Exchange timestamp: Bytes 8–15

int64 · 8 bytes

Price: Bytes 16–23

float64 · 8 bytes

Quantity: Bytes 24–31

float64 · 8 bytes

Time irregularity across instruments

Once we decide how to tokenise the order book, we can represent a sequence as a matrix:
$X \in \mathbb{R}^{L \times d_{\text{input}}}$,
where L is the number of events or snapshots in the sequence; for a discrete tokenizer we'd have a vocabulary of size V. For e.g. an order-book snapshot, we'd have 40 features.

Suppose we want to train for multiple instruments.

If we want a single model across multiple instruments, we can introduce a learned instrument embedding:
$h_{i,k} = f(x_{i,k}) + e_{\text{instrument},i}$

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But this doesn't solve all of our problems: event time is not clock time
Suppose L = 50,000 events. For a very liquid stock like NVDA this may be 1 minute, while it can be 3 days for a less liquid stock.

- We could feed also a time and position embedding:$h_{i,k} = f(x_{i,k}) + e_{\text{instrument},i} + e_{\text{position},k} + e_{\text{time},i,k}$.
- We could instead also **bin** onto equal time bins

Long context challenges

Imagine a model trained to predict the **10-minute** movement in NVDA. It may need to understand the immediate book dynamics, but also the preceding **hours** of trading.

If 50,000 observations cover only a minute, extending the context to an hour would require 3 million events at the same hypothetical rate.

Standard self-attention has quadratic computational complexity in sequence length:
Attention complexity $= O(L^2 d)$

Long context challenges

Luckily, LLMs have faced the same challenges when extending to longer contexts, e.g. due to agentic trajectories and reasoning chains.

Architecture	Main idea	Sequence complexity
Sliding-window attention	Attend only to the previous \(w\) tokens	O(Lw)
Sparse attention	Attend to \(k\) selected tokens or blocks	O(Lk)
Linear attention	Replace pairwise attention with an accumulated state	O(L)
Mamba / Mamba-2	Selective state-space recurrence	O(L)
Kimi Delta Attention (KDA)	Gated DeltaNet with finer-grained memory control	O(L)

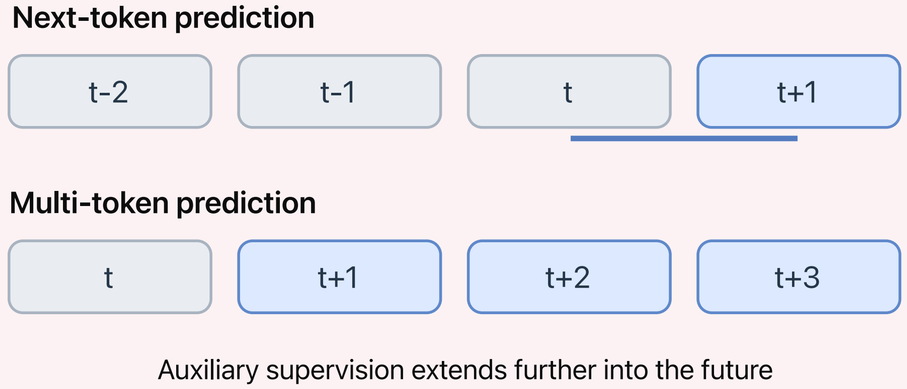
Target definition

Next-token prediction:
In LLMs, the target is quite logical:

$$\mathscr{L}_{\text{NTP}} = -\sum_{t} \log p_\theta(x_{t+1} \mid x_{\leq t})$$

At each position, the model predicts the next token.

Newer models like DeepSeek extend this with e.g. MTP to predict several tokens into the future, but the logic remains the same.

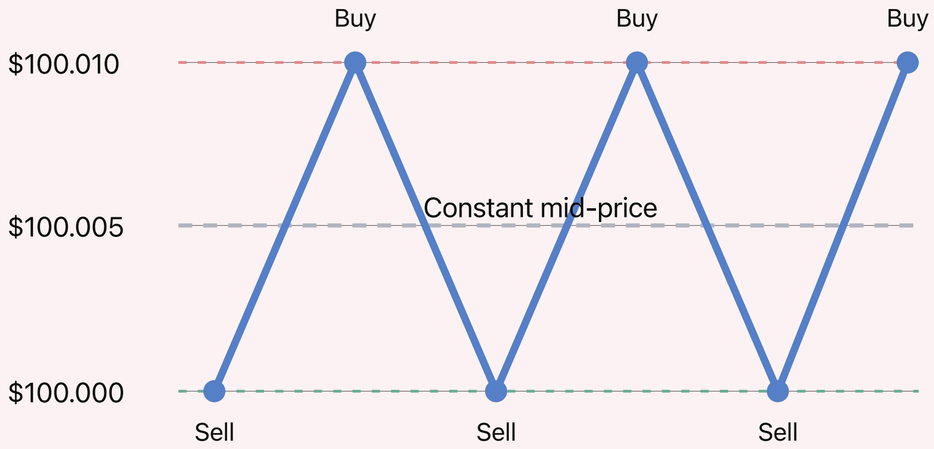


Auxiliary supervision extends further into the future

Target definition

Next-token prediction:
But in finance, on very short scales, e.g. next event, we have many **mechanic effects.**

E.g. the bid-ask bounce arising because buyers and sellers alternate.



A naive model trained on transaction prices sees:
100.00 → 100.01 → 100.00 → 100.01 → ⋯
It sees pronounced mean reversion, but this is not tradeable.
E.g. Suppose the last transaction occurred at $100.01 (the ask), and your model predicts the next transaction will occur at $100.00 (the bid). You might think you can profit by selling at $100.01 and buying back at $100.00, earning $0.01 per share. But $100.01 is the price at which someone just bought, not the price at which you can immediately sell.

Other things: for liquid instruments liquidity gets replenished often after being removed. But may be too fast to trade.

Loss: point-wise or distribution-based?

We typically care about the Gain: $R_t = p_t \, r_{t,t+H}$

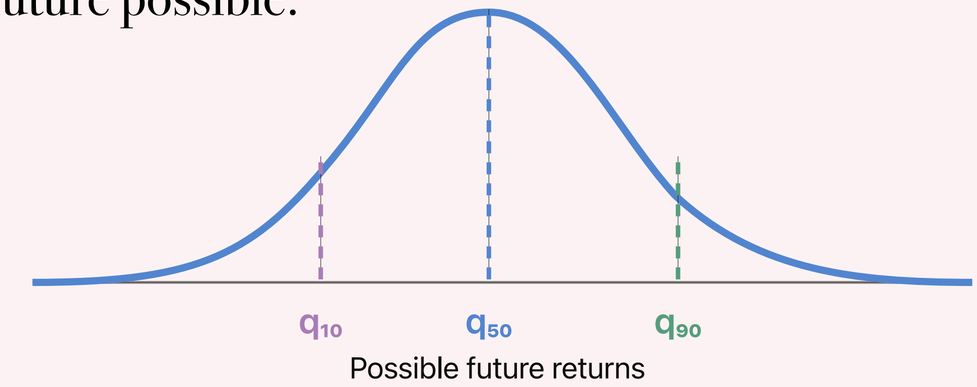
Mean Squared Error:
$\mathscr{L}_{\text{MSE}} = \mathbb{E}[(r_t - p_t)^2] = \mathbb{E}[r_t^2] - 2\mathbb{E}[r_t p_t] + \mathbb{E}[p_t^2] = -2R_t + \mathbb{E}[p_t^2] + C$

Loss: point-wise or distribution-based?

But: the book is not deterministic: there is not only one single future possible.

Quantile loss: the final output head maps into Q quantiles:

$$\mathscr{L}_\tau(y, \hat{q}) = \begin{cases} \tau(y - \hat{q}), & y \geq \hat{q} \\ (1 - \tau)(\hat{q} - y), & y < \hat{q} \end{cases}$$



Diffusion loss: Standard quantile regression learns marginal conditional distributions, whereas a multivariate diffusion model can learn the full joint conditional distribution across features and future time steps.

Source: Backhouse et al. (2025), Painting the Market: Generative Diffusion Models for Financial Limit Order Book Simulation and Forecasting

$$\mathscr{L}_{\text{diff}} = \mathbb{E}_{x,\epsilon,s}\left[\|\epsilon - \epsilon_\theta(x_s, s, c)\|^2\right]$$

Correlated time series

Imagine a market maker providing liquidity in NVDA, currently trading at $100.
The market maker quotes: **Bid: volume at $99.99. Ask: volume at $100.01.**
Then **a large investor sells 10,000 NVDA shares.**
The market maker buys them at $99.99, absorbing the sell pressure. It is now **long 10,000 NVDA shares**. If NVDA falls by $1, it loses $10,000 on that position.

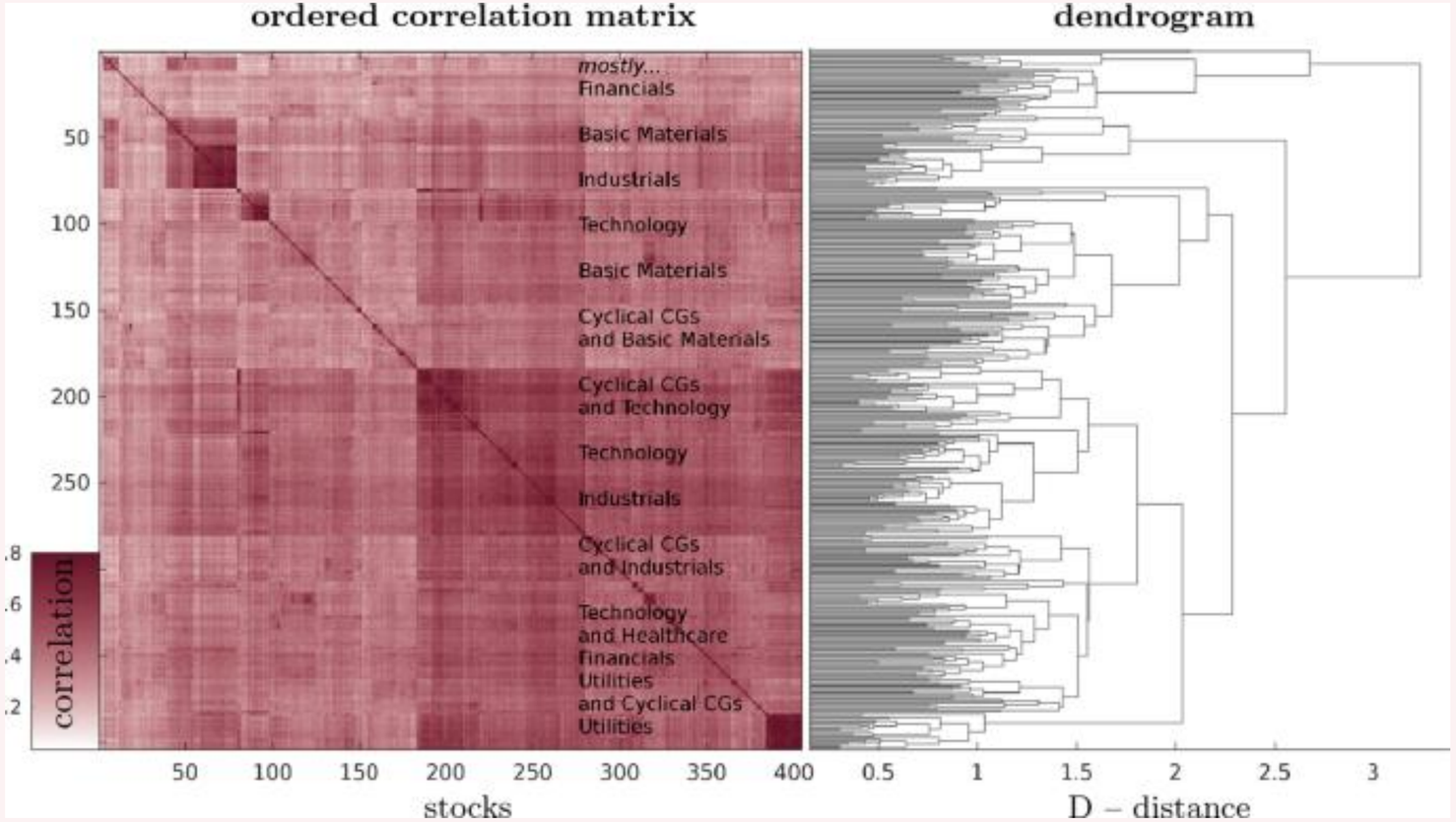
The market maker needs to reduce its risk.
He can **hedge using another instrument**: for example, short the semiconductor ETF SMH, which has exposure to NVDA and related semiconductor companies.

This leads to cross-instrument correlations.

Correlated time series

Source: Raddant & Di Matteo (2023), A Look at Financial Dependencies by Means of Econophysics and Financial Economics

Stocks exhibit clusters of correlated returns. Companies in similar sectors frequently move together, producing visible blocks in the correlation matrix.



Cross-covariate attention

Suppose that we would like to process **D time-series** of embedding size d_model: e.g. the stock we are trying to predict, and several correlated stocks as covariates. Recent time series foundation models, e.g. Timestral and TFC's to use:

Hidden representation

D × L × d_model

Alternate:

Covariate attention

mix along D, for each time patch

highlighted row = one time patch
attention moves across variates

Temporal attention

mix along L, for each variate

highlighted column = one variate
attention moves across time

& the MLP over d_model

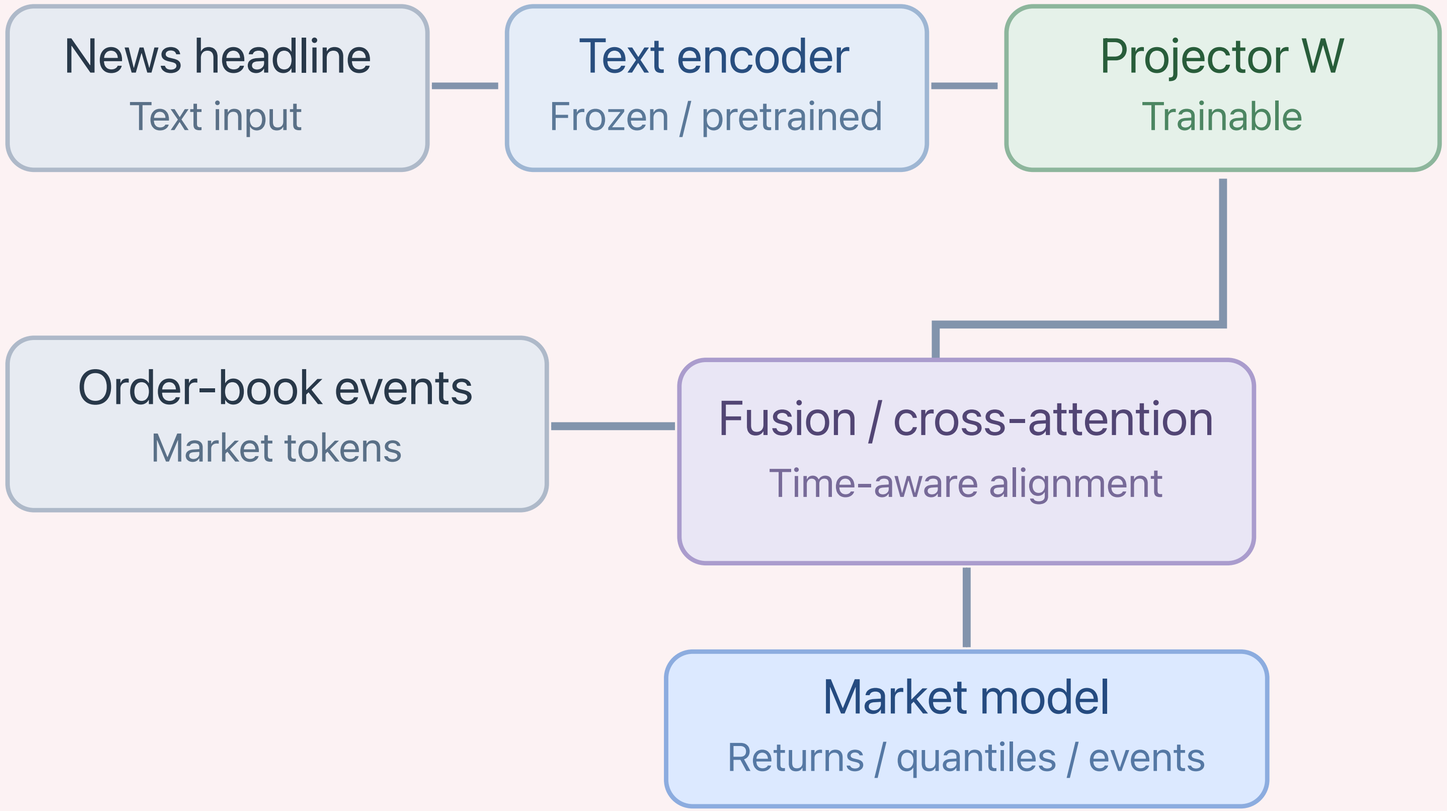
Auxiliary information

Imagine NVDA releases an earnings report.

The order book reflects rapidly changing prices and liquidity, but the text itself contains information about revenue, guidance, margins, product demand and management expectations.

- A pure order-book model only sees how market participants react.
- A multimodal model could also receive a representation of the underlying information.

Inspiration: vision-language models such as LLaVA: aligning different modalities with a learned projection

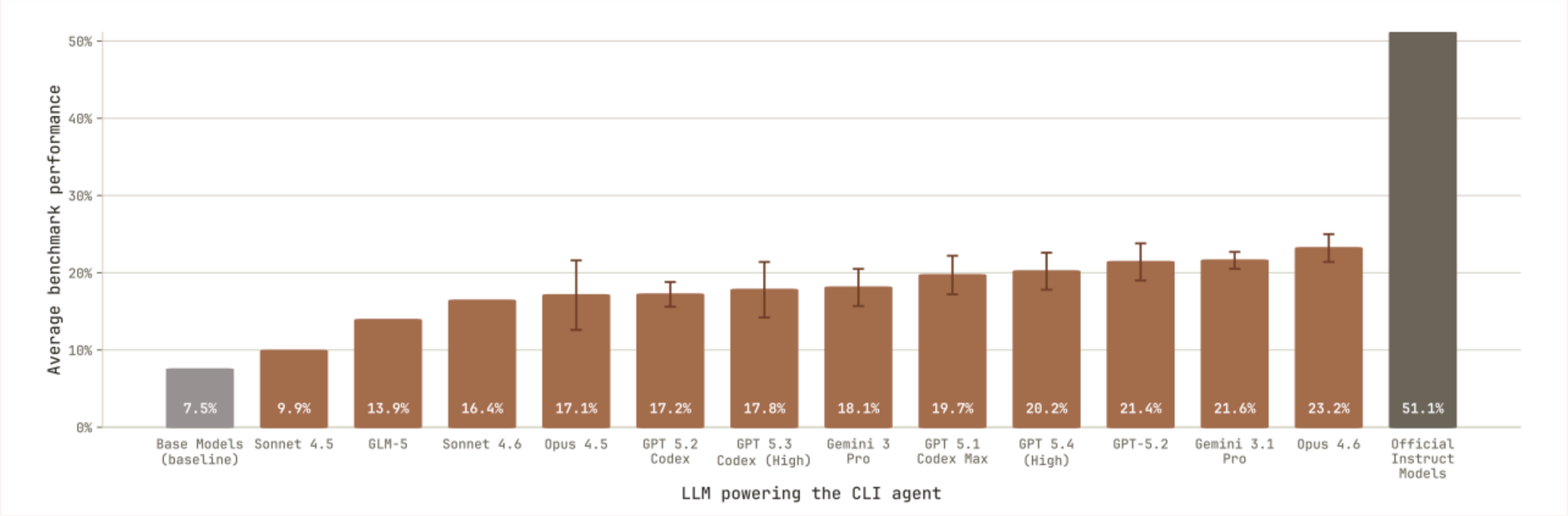


Auto-Research in LLMs

PostTrainBench

Source: Rank et al. (2026), PostTrainBench: Can LLM Agents Automate LLM Post-Training?

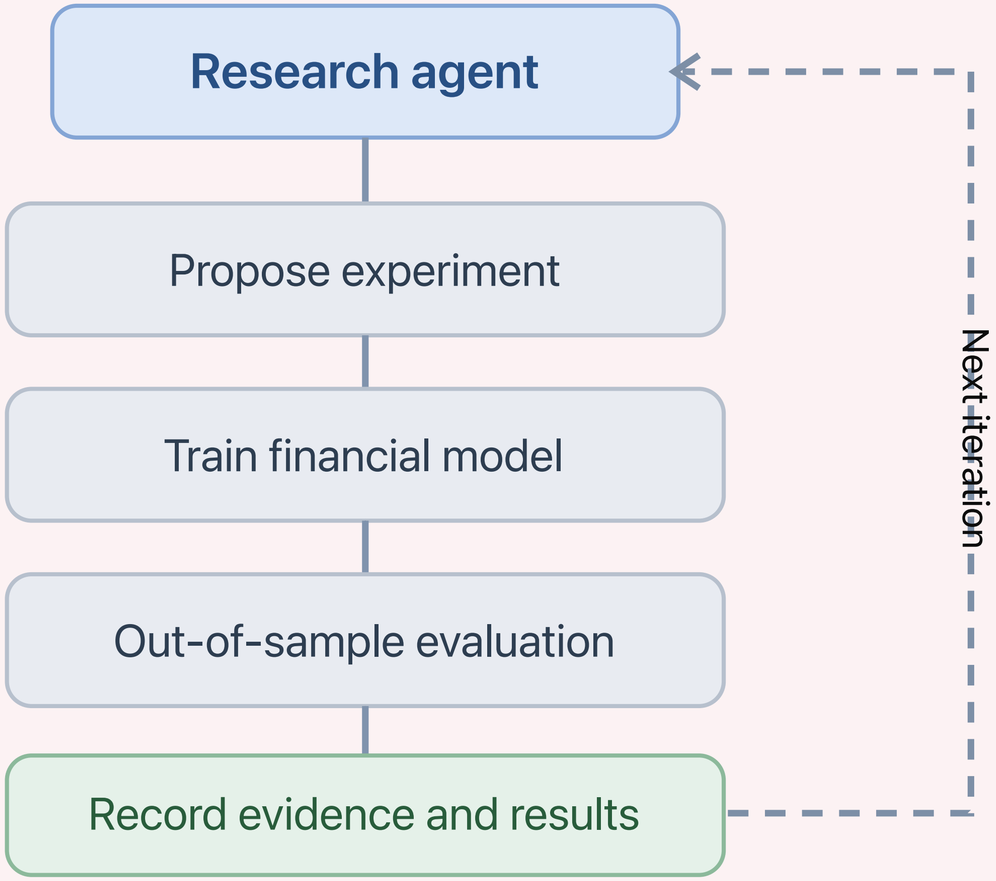
An agent is given a base LLM, a target benchmark and a restricted experimental environment. It must decide what to try, find or prepare training data, modify training code, run experiments and submit an improved model.



An auto-research harness

For example, an agent might initially train a 50-million-parameter model to predict event tokens.

After seeing weak downstream return performance, it could test quantile-based targets, change the volume encoding, introduce explicit time embeddings, compare event-based and clock-time contexts, or add cross-instrument features.



An auto-research harness

